

Full-Stack Development Task for Interns

Objective

Build a fully functional eCommerce web application based on the [Ecommerce Web Design Figma template](#). This task requires completing both the desktop and responsive (mobile view) versions with backend integration for dynamic content.

Tasks Overview

The project is divided into weekly milestones, covering frontend development, backend integration, and final deployment.

Week 1: Project Setup and Static Frontend Development

Goal

Set up the project environment and implement the static frontend for desktop and mobile views.

Tasks

1. Set up the development environment:
 - Install and configure **Node.js**, **React** (frontend), and **Express.js** (backend).
2. Clone the [Figma design template](#) into your project.
3. Develop the following static pages for desktop and mobile views:
 - **Home Page**
 - **Product Listing Page**
 - **Product Details Page**
 - **Cart Page**
4. Ensure the designs are fully responsive using **CSS Grid**, **Flexbox**, or any CSS framework like **TailwindCSS** or **Bootstrap**.

Deliverables

- Responsive frontend for desktop and mobile views.
- Pages match the Figma design.
- Code committed to a GitHub repository named [ecommerce-fullstack-design](#).

Week 2: Backend Setup and Dynamic Integration

Goal

Set up the backend and connect it to the frontend for dynamic content rendering.

Tasks

1. Set up a **MongoDB** (or Firebase Realtime Database) for storing product data.
2. Create a **Node.js** and **Express.js** backend with:
 - **API endpoints** for CRUD operations (Create, Read, Update, Delete) for products.
 - A **Products Collection** with attributes:
 - **id, name, price, image, description, category, stock.**
3. Populate the database with sample product data.
4. Fetch data dynamically from the backend to render the following pages:
 - **Home Page:** Display featured products.
 - **Product Listing Page:** Fetch and display all products in a grid layout.
 - **Product Details Page:** Display details for a selected product.
 - **Cart Page:** Add products to a cart and display selected items.
5. Implement a **search bar** to filter products by name or category.

Deliverables

- Functional backend with APIs for product management.
- Fully integrated frontend displaying dynamic data.
- Code committed to the GitHub repository.

Week 3: Additional Features and Final Deployment

Goal

Add essential eCommerce features, complete responsive testing, and deploy the application.

Tasks

1. **User Authentication:**

Use **JWT** or **Firestore Auth** for login and signup functionality. ○ Restrict access to admin pages (e.g., product management).

2. **Cart Management:**

Enable users to add/remove products from the cart.

- Persist cart data locally using **localStorage** or a backend database.

3. **Admin Panel:**

- Create a simple admin interface to add/edit/delete products.
- Use protected routes for the admin panel.

4. **Responsive Testing:**

Test and fix any responsive issues for desktop and mobile views.

5. **Deployment:**

Deploy the project to **Render**, **Vercel**, or **Heroku**.

Deliverables

- User authentication implemented and tested.
- Cart functionality working for both desktop and mobile.
- Admin panel integrated with CRUD operations.
- Deployed application with live URL.
- Final code committed to the GitHub repository.

Tools and Technologies

- **Frontend:** React.js, CSS3, TailwindCSS/Bootstrap (optional).
- **Backend:** Node.js, Express.js.
- **Database:** MongoDB/Firebase Realtime Database.
- **Version Control:** GitHub.
- **Deployment:** Render, Vercel, or Heroku.
- OR YOU CAN USE TECHNOLOGY STACK OF YOUR OWN CHOICE

Submission Instructions

1. Create a GitHub repository named [ecommerce-fullstack-design](#).
2. Push your code weekly based on deliverables.
3. Share the repository link and deployed URL with your supervisor for review.
4. Record a video of your project while running and performing operations and upload it on google classroom.(on **16th February, 2026**)

Evaluation Criteria

Functionality

- Fully functional frontend and backend with dynamic content.
- Responsive design for desktop and mobile.

Code Quality

- Clean and readable code with proper comments.
- • Well-organized folder structure.

Database Design

- Efficient and scalable database schema.

Timely Submission

- Weekly milestones completed and pushed to GitHub on time.

Deadline: 15th May, 2026

(You will have to submit all of these 3 tasks collectively on your consoles)

Good Luck!

Ensure your project is well-documented, responsive, and fully functional to deliver an excellent user experience.